



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Disaster Response & Resource Allocation System	
S. No.	University ID	Name
1	2420030638	Rayapureddi Rishi SriCharan
2	2420030161	Yennam Sesank Reddy
3	2420030282	Harsha Koganti
4	2420030276	Aluru Sri Tejaswi Priyank
5	24200907	Harshath Ramannagari
Name of the Guide		Anugu Swapna

Abstract

Natural disasters such as floods, earthquakes, cyclones, and wildfires pose significant challenges to emergency response agencies due to rapidly changing environmental conditions, resource limitations, and the need for timely decision-making.

Traditional disaster management systems often rely on static workflows and manual coordination, making it difficult to respond effectively to dynamic disaster scenarios.

To address these challenges, this project proposes the development of an Adaptive AI-Powered Disaster Response & Resource Allocation System with Real-Time Disaster Simulation.

The proposed system combines Artificial Intelligence (AI), real-time simulation, and adaptive software engineering principles to create an intelligent disaster management platform capable of monitoring evolving disaster situations, analyzing their impact, and dynamically allocating emergency resources.

The system simulates realistic disaster scenarios such as floods, earthquakes, and wildfires on a virtual city map, enabling users to observe how emergency services respond as conditions change over time.

Based on factors including disaster severity, affected population, available rescue teams, hospital capacity, road accessibility, and shelter occupancy, the AI engine continuously prioritizes incidents, recommends optimal rescue strategies, reallocates resources, and updates evacuation routes in real time.

A key feature of the project is its self-adaptive decision-making capability, implemented using the Monitor–Analyze–Plan–Execute over a Knowledge Base (MAPE-K) architectural model.

This enables the system to automatically react to changing environmental conditions without requiring manual intervention.

The platform also provides an interactive dashboard displaying live disaster progression, emergency resource utilization, rescue operations, predictive analytics, and performance metrics, offering users a comprehensive view of disaster response activities.

Beyond serving as a disaster management application, the system functions as a simulation environment for evaluating emergency response strategies under different disaster conditions.

The proposed solution demonstrates the integration of adaptive software engineering, artificial intelligence, event-driven simulation, and data visualization to improve disaster preparedness, optimize resource utilization, and support informed decision-making.

The project aims to provide an innovative, scalable, and research-oriented framework that can be extended for smart city applications and future emergency management systems.

Signature of the Guide

HOD